The Minimum Self-Consistent Relational Structure

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1 Definitions

Definition 1.1 (Primitive space)**.** A *primitive space* is a finite-dimensional complex vector space S with $\dim_{\mathbb{C}}(S) \geq 2$. The elements of S are called *primitive entities*. The condition $\dim(S) \geq 2$ encodes the requirement that at least two distinguishable entities exist: below two, there is no relation, hence no structure.

Definition 1.2 (Relational structure)**.** A *relational structure* on a primitive space S is a triple (S, V, π) where:

1. V is a finite-dimensional complex vector space (the *state space*),
2. $\pi\colon S \times S \to V$ is the canonical bilinear map $\pi(u, v) = u \otimes v$, identifying V with $S \otimes S$.

The bilinear map π encodes the assertion that every state arises from the interaction (tensor product) of primitive entities.

Definition 1.3 (Dynamical and substrate decomposition)**.** The tensor product $S \otimes S$ decomposes under the symmetric group $\mathfrak{S}_2$ as

$$S \otimes S \;=\; \mathrm{Sym}^2(S) \;\oplus\; \Lambda^2(S).$$

The *dynamical algebra* is $\mathrm{Sym}^2(S)$: the symmetric interactions, which carry the $\mathrm{SL}(S)$-adjoint representation and encode causal structure (see Corollary 4.1). The *substrate* is $\Lambda^2(S)$: the antisymmetric part, which carries the invariant volume forms and provides the medium in which relations are embedded.

Definition 1.4 (Self-consistency)**.** A relational structure (S, V, π) is *self-consistent* if $V = S \otimes S$—that is, the state space is neither more nor less than the bilinear interactions of the primitives.

Remark 1.5 (Why $V = S \otimes S$)*.* Self-consistency demands that the state space be generated by the primitive entities under their mutual interaction. The tensor product $S \otimes S$ is the universal bilinear construction: it is the smallest space through which every bilinear map on S factors. To take $V \subsetneq S \otimes S$ would discard bilinear interactions, breaking universality. To take $V \supsetneq S \otimes S$ would require structure not generated by the primitives, violating self-containment.

2 Axioms

Axiom 1 (Distinguishability)**.** The primitive space S is a finite-dimensional complex vector space with $\dim_{\mathbb{C}}(S) \geq 2$.

Axiom 2 (Self-consistency)**.** The state space is $V = S \otimes S$.

Axiom 3 (Unique substrate)**.** $\dim_{\mathbb{C}}(\Lambda^2(S)) = 1$.

Remark 2.1 (On the axiom count). These three axioms are not independent. Axiom 3 alone forces $\dim(S) = 2$, which then satisfies Axiom 1. The logical content is:

1. $V = S \otimes S$ (self-consistency),
2. $\dim(\Lambda^2(S)) = 1$ (unique substrate),

from which everything follows. Axiom 1 is retained for conceptual clarity: it makes explicit the minimal relational requirement before the substrate constraint is imposed.

3 The Main Theorem

Theorem 3.1 (Minimum self-consistent relational structure)**.** *Under Axioms 1–3:*

(i) $\dim(S) = 2$, *so* $S \cong \mathbb{C}^2$.

(ii) $\dim(V) = 4$, *so* $V \cong \mathbb{C}^4$.

(iii) Setting $H := \dim(\mathrm{Sym}^2(S))$, *we have* $H = 3$ *and the self-consistency identity*

$$(H-1)^2 = H + 1 \tag{1}$$

is satisfied.

Proof. Let $n = \dim_{\mathbb{C}}(S)$.

(i) The exterior power $\Lambda^2(S)$ has dimension $\binom{n}{2} = n(n-1)/2$. Axiom 3 requires $n(n-1)/2 = 1$, i.e., $n(n-1) = 2$. This is a quadratic in n with roots $n = 2$ and $n = -1$. Since n is a positive integer, $n = 2$.

(ii) By Axiom 2, $\dim(V) = \dim(S \otimes S) = n^2 = 4$.

(iii) $\dim(\mathrm{Sym}^2(S)) = \binom{n+1}{2} = n(n+1)/2$. At $n = 2$: $H = 2 \cdot 3/2 = 3$. The decomposition $\dim(V) = \dim(\mathrm{Sym}^2(S)) + \dim(\Lambda^2(S))$ gives $4 = 3 + 1$, which is $H + 1 = 4$. Meanwhile, $(H-1)^2 = (3-1)^2 = 4$. Hence $(H-1)^2 = H + 1$.

Alternatively, equation (1) is derived from the dimension identity. We have $\dim(V) = n^2$ and $\dim(\mathrm{Sym}^2(S)) = n(n+1)/2 = H$, $\dim(\Lambda^2(S)) = n(n-1)/2$. The decomposition $n^2 = H + n(n-1)/2$ gives $H = n^2 - n(n-1)/2 = n(n+1)/2$. The constraint $\dim(\Lambda^2(S)) = 1$ means $n(n-1)/2 = 1$, so $\dim(V) = H + 1$. Now $(H-1)^2 = \left(\frac{n(n+1)}{2} - 1\right)^2$. At $n = 2$ (the unique solution): $(3-1)^2 = 4 = 3 + 1 = H + 1$. To see that this is the *content* of the unique substrate axiom in the self-consistency equation: substituting $\dim(V) = H + 1$ into the identity $\dim(V) = (H-1)^2$ would require $(H-1)^2 = H + 1$, i.e., $H^2 - 3H = 0$, giving $H = 3$ as the unique positive solution. □

Remark 3.2 (The equation $(H-1)^2 = H + 1$). In the present context, equation (1) is a consequence of the dimension identity $\dim(V) = H + 1$ forced by $\dim(\Lambda^2(S)) = 1$. It acquires a second, independent meaning when read in the tangent bundle setting (Corollary 4.5): the left side $(H-1)^2$ counts the dimension of traceless symmetric bilinear forms on $H - 1$ independent directions—the number of independent bilinear products the dynamical algebra can produce—and the right side $H + 1 = \dim(V)$ is the dimension of the full state space. The equation then says: the self-interaction of the dynamical algebra *exactly fills* the state space. That the same equation arises from two different starting points (unique substrate, and tangent bundle self-consistency) is the mathematical content of the uniqueness.

4 Corollaries

Corollary 4.1 (Dynamical algebra is $\mathfrak{sl}(2,\mathbb{C})$)**.** $\mathrm{Sym}^2(\mathbb{C}^2)$ *is isomorphic, as an* $\mathrm{SL}(2,\mathbb{C})$*-representation, to the adjoint representation* $\mathfrak{sl}(2,\mathbb{C})$*.*

Proof. The natural $\mathrm{SL}(2,\mathbb{C})$-action on $S = \mathbb{C}^2$ induces an action on $\mathrm{Sym}^2(S)$. The adjoint representation of $\mathrm{SL}(2,\mathbb{C})$ on its Lie algebra $\mathfrak{sl}(2,\mathbb{C})$ has dimension 3. Both are irreducible representations of dimension 3 of $\mathrm{SL}(2,\mathbb{C})$ (the spin-1 representation), hence isomorphic.

Explicitly: let $\{e_0, e_1\}$ be a basis of S. The symmetric products e_0^2, e_0e_1, e_1^2 form a basis of $\mathrm{Sym}^2(S)$. Under the standard identification with homogeneous polynomials of degree 2 in two variables, the $\mathrm{SL}(2,\mathbb{C})$-action by linear substitution gives the spin-1 (adjoint) representation. □

Corollary 4.2 (Substrate is the symplectic form)**.** $\Lambda^2(\mathbb{C}^2) \cong \mathbb{C}$ *carries the unique (up to scale)* $\mathrm{SL}(2,\mathbb{C})$*-invariant antisymmetric bilinear form* $\varepsilon\colon S \times S \to \mathbb{C}$*. This is the spinor metric (the* ε*-spinor of Penrose–Rindler).*

Proof. $\Lambda^2(\mathbb{C}^2)$ is one-dimensional, hence carries the trivial $\mathrm{SL}(2,\mathbb{C})$-representation (since $\mathrm{SL}(2,\mathbb{C})$ preserves the volume form by definition). The generator $e_0 \wedge e_1$ defines ε_{AB} with $\varepsilon_{01} = -\varepsilon_{10} = 1$, the standard symplectic form on $\mathbb{C}^2$. That $\mathrm{SL}(2,\mathbb{C})$ is the group preserving ε (up to sign) confirms the identification. □

Corollary 4.3 (Twistor space)**.** *The projectivization of the state space is* $\mathbb{P}(V) = \mathbb{CP}^3$*, which is Penrose's twistor space for conformally compactified Minkowski space.*

Proof. $V = \mathbb{C}^4$, so $\mathbb{P}(V) = \mathbb{CP}^3$. The identification with Penrose's twistor space $\mathbb{PT}$ follows from the $\mathrm{SL}(2,\mathbb{C})$-decomposition. The state space $V = S \otimes S = \mathbb{C}^2 \otimes \mathbb{C}^2$ carries the natural $\mathrm{SL}(2,\mathbb{C}) \times \mathrm{SL}(2,\mathbb{C})$ action. In the twistor formalism, a twistor $Z^\alpha = (\omega^A, \pi_{A'})$ decomposes under the spin group $\mathrm{SL}(2,\mathbb{C})_L \times \mathrm{SL}(2,\mathbb{C})_R$ of the conformal group, with $\omega^A \in S_L$ and $\pi_{A'} \in S_R$. The conformal group of compactified Minkowski space is $\mathrm{SL}(4,\mathbb{C})/\mathbb{Z}_4 \cong \mathrm{SO}(6,\mathbb{C})$, acting naturally on $\mathbb{C}^4$. The twistor fibration $\mathbb{CP}^3 \to S^4$ is the standard Penrose fibration. □

Corollary 4.4 (Holomorphic sections of the tangent bundle)**.** $\mathrm{Sym}^2(\mathbb{C}^2) \cong H^0(\mathbb{CP}^1, \mathcal{O}(2))$*: the dynamical algebra is the space of holomorphic sections of the tangent bundle of the Riemann sphere.*

Proof. The tangent bundle of $\mathbb{CP}^1$ is the line bundle $\mathcal{O}(2)$. Its holomorphic sections are homogeneous polynomials of degree 2 in the homogeneous coordinates $[\pi_0 : \pi_1]$ of $\mathbb{CP}^1$:

$$\eta(\pi) = s_0\,\pi_0^2 + s_1\,\pi_0\pi_1 + s_2\,\pi_1^2, \qquad (s_0, s_1, s_2) \in \mathbb{C}^3.$$

This is precisely $\mathrm{Sym}^2(S^\vee) \cong \mathrm{Sym}^2(S)$ (using the symplectic form ε to identify $S \cong S^\vee$). Hence $\dim H^0(\mathbb{CP}^1, \mathcal{O}(2)) = 3 = H$. □

Corollary 4.5 (The quartic factorization)**.** $\mathbb{CP}^1$ *is the unique compact complex projective space* $\mathbb{CP}^n$ *(*$n \geq 1$*) whose tangent bundle satisfies the self-consistency equation.*

Proof. For $\mathbb{CP}^n$, $\dim H^0(T\mathbb{CP}^n) = (n+1)^2 - 1$ (the generators of $\mathrm{PGL}(n+1,\mathbb{C})$). Substituting $H = (n+1)^2 - 1$ into $(H-1)^2 = H + 1$:

$$((n+1)^2 - 2)^2 = (n+1)^2.$$

Expanding: $n^4 + 4n^3 + n^2 - 6n = 0$, which factors completely over $\mathbb{Z}$:

$$n(n-1)(n+2)(n+3) = 0.$$

Roots: $n \in \{0, 1, -2, -3\}$. The unique positive integer is $n = 1$. □

5 Critical Assessment

5.1 Status of Axiom 2: $V = S \otimes S$

This is the strongest axiom and the one most open to challenge. The claim is that the state space equals the full tensor product of the primitive space with itself. Three points bear scrutiny.

Why not $V = \mathrm{Sym}^2(S)$? One might argue that only the symmetric part—the "observable" interactions—should constitute the state space, discarding the antisymmetric part as unphysical. This would give $V = \mathrm{Sym}^2(S)$, $\dim(V) = n(n+1)/2$, and no substrate at all. The counter-argument is structural: the tensor product $S \otimes S$ is the *universal* bilinear construction. Every bilinear map $S \times S \to W$ factors through $S \otimes S$, not through $\mathrm{Sym}^2(S)$. To restrict to $\mathrm{Sym}^2(S)$ is to impose a symmetry condition *before* asking what the structure requires—it is an additional axiom, not a simplification. The full tensor product is the canonical, assumption-free choice. Furthermore, discarding $\Lambda^2(S)$ eliminates the symplectic structure that defines the spinor metric, destroying the very object (the ε-spinor) that makes S a spin space.

Why not $V = S \otimes S'$ for $S' \neq S$? Self-consistency requires that the state space be generated by interactions of the *same* primitive entities with themselves. Introducing a second, independent primitive space S' would add external structure not determined by S, violating the closure condition.

Why not higher tensor powers? One could consider $V = S^{\otimes k}$ for $k \geq 3$. The bilinear requirement ($k = 2$) is the minimum: a single interaction between two entities. Higher tensor powers describe multi-party interactions and are *derived* from the bilinear structure, not primitive. The tensor product $S \otimes S$ is the irreducible case.

5.2 Status of Axiom 3: $\dim(\Lambda^2(S)) = 1$

This axiom encodes uniqueness of the substrate. It is the sharpest constraint and admits two readings:

Reading 1: Minimality. The substrate is the medium through which relations are embedded. Multiple independent substrates ($\dim(\Lambda^2(S)) > 1$) would require a meta-substrate to relate them, producing infinite regress. One substrate is the minimum that avoids this regress. Under this reading, Axiom 3 is a minimality condition, not a logical necessity.

Reading 2: Non-degeneracy. For S to be a spin space, it must carry a non-degenerate symplectic form $\varepsilon \in \Lambda^2(S^\vee)$. Non-degeneracy requires $\Lambda^2(S)$ to be one-dimensional (so that the symplectic form is unique up to scale and cannot degenerate into a lower-rank object). For $\dim(S) \geq 3$, $\Lambda^2(S)$ has dimension ≥ 3, and the "symplectic form" is no longer unique—there is a moduli space of symplectic structures. The condition $\dim(\Lambda^2(S)) = 1$ is equivalent to requiring that the antisymmetric structure be *rigid*: no continuous deformations, no moduli, no choices.

Both readings lead to the same constraint, but neither is a logical theorem—each rests on an interpretive commitment (no regress, or rigidity of the antisymmetric structure). This is the weakest point of the axiom system.

5.3 Alternative axiom systems

Dropping Axiom 3. Without the unique substrate condition, $\dim(S)$ is unconstrained (subject to $\dim(S) \geq 2$). For $\dim(S) = 3$: $V = \mathbb{C}^9$, $\mathrm{Sym}^2(S) = \mathbb{C}^6$, $\Lambda^2(S) = \mathbb{C}^3$. The self-consistency equation $(H-1)^2 = H+1$ with $H = 6$ gives $25 \neq 7$: it fails. For general n, setting $H = n(n+1)/2$ and $\dim(V) = n^2$, the condition $(H-1)^2 = \dim(V)$ becomes $\left(\frac{n(n+1)}{2} - 1\right)^2 = n^2$, which gives $n = 2$ as the unique positive integer solution with $n \geq 2$. Taking the positive square root: $\frac{n(n+1)}{2} - 1 = n$, i.e., $n^2 + n - 2 = 2n$, i.e., $n^2 - n - 2 = 0 = (n-2)(n+1) = 0$, giving $n = 2$. (The negative root $\frac{n(n+1)}{2} - 1 = -n$ gives $n^2 + 3n - 2 = 0$, which has no positive integer solution.)

Direct verification: at $n = 2$, $\left(\frac{2\cdot 3}{2} - 1\right)^2 = (3-1)^2 = 4 = 2^2$, confirmed. This shows that even replacing Axiom 3 with the self-consistency equation $(H-1)^2 = \dim(V)$ directly, $n = 2$ is the unique solution with $n \geq 2$. In other words, the self-consistency equation and the unique substrate condition are *equivalent* in this setting.

Real instead of complex. If S is a real vector space, $\Lambda^2(S)$ carries an orthogonal (not symplectic) structure. The condition $\dim(\Lambda^2(S)) = 1$ still gives $\dim(S) = 2$, yielding $V = \mathbb{R}^4$ and $H = 3$. But the dynamical algebra is $\mathfrak{sl}(2, \mathbb{R})$, the split real form, whose Killing form has signature $(1, 2)$. The absence of complex structure means $\mathbb{R}^4$ does not projectivize to a complex manifold, and the holomorphic geometry essential to twistor theory is unavailable. The complex structure on S is necessary for the correct signature and for the holomorphic geometry of twistor theory. This is consistent with Theorem 9 of the main paper (uniqueness of $\mathbb{C}$ among normed division algebras with an ordering-sensitive encoding).

Quaternionic. If S is a right $\mathbb{H}$-module with $\dim_{\mathbb{H}}(S) = 1$ (the quaternionic analogue of $\mathbb{C}^2$), then $S \otimes_{\mathbb{H}} S$ is not well-defined in the same sense (the tensor product over $\mathbb{H}$ does not yield a free $\mathbb{H}$-module in general because $\mathbb{H}$ is non-commutative). Treating S as $\mathbb{C}^4$ with a quaternionic structure j, the self-consistency analysis leads to $\dim(\Lambda^2_{\mathbb{C}}(S)) = 6$, violating the unique substrate condition.

5.4 Summary of logical status

Axiom	**Status**	**Weakest point**
A1: $\dim(S) \geq 2$	Definitional	"Two entities" is a starting premise
A2: $V = S \otimes S$	Structural	Why not a subspace? (Answer: universality)
A3: $\dim(\Lambda^2(S)) = 1$	Minimality	"One substrate" requires interpretation

The strongest result is that A2 and A3 are not independent in their consequences: the self-consistency equation $(H-1)^2 = \dim(V)$ with $V = S \otimes S$ has $n = 2$ as its unique solution with $n \geq 2$, and at $n = 2$ one automatically gets $\dim(\Lambda^2(S)) = 1$. Conversely, $\dim(\Lambda^2(S)) = 1$ forces $n = 2$, which then satisfies the self-consistency equation. The two constraints are equivalent. This is either a strength (the axiom system is overdetermined and consistent) or a weakness (one axiom is redundant), depending on perspective.

Remark (What is proven vs. assumed). *Proven:* Given a complex vector space S with $\dim(S) = 2$, the tensor product $V = S \otimes S = \mathbb{C}^4$ decomposes as $\mathrm{Sym}^2(S) \oplus \Lambda^2(S) = \mathbb{C}^3 \oplus \mathbb{C}$, and this is the unique dimension satisfying $(H-1)^2 = H + 1$. The corollaries (adjoint representation, symplectic form, twistor space, tangent bundle sections) are standard results in representation theory and algebraic geometry.

Assumed: That the state space of a self-consistent relational structure must be $S \otimes S$ (not some other functor of S), and that the substrate must be unique. These are interpretive commitments grounded in universality and minimality, not logical necessities. The mathematical content is rigorous; the philosophical motivation for the axioms is, as with all axiom systems, a matter of judgment.